

User-Defined Behavioral State Labeler — Design Mockups

visdetect / BG_046 · design session 2026-06-10

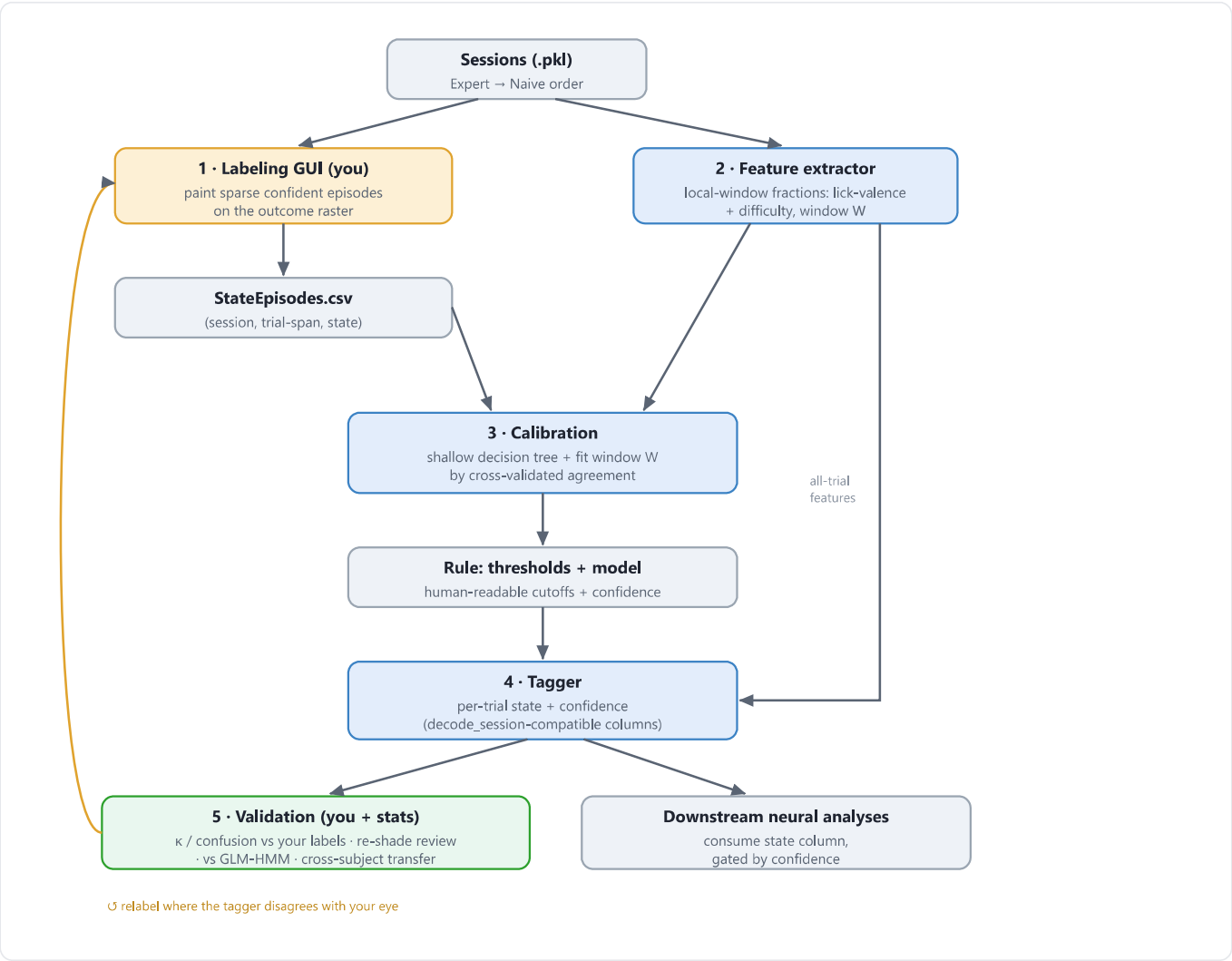
The problem. The GLM-HMM (Ashwood) infers *latent* states by fitting GLM weights that maximize trial-by-trial lick likelihood. Its boundaries land wherever the likelihood surface puts them — there is no place for the experimenter's judgment about where a state begins or ends. So the states often don't match where one would draw them by eye.

The plan (human-in-the-loop, data-driven).

1. **Label.** Read the trial-by-trial *outcome raster* and paint only the episodes you're confident about (sparse). Start with **Expert** sessions where states are clearest, then work backward toward Naive.
2. **Calibrate.** Turn those labels into an **interpretable threshold rule** on local outcome composition (FA / miss / hit fractions, difficulty-aware), and *fit the window width* to match your eye. (A soft probabilistic classifier is kept as a documented future comparison.)
3. **Tag.** Apply the rule to every session, with a per-trial **confidence** so ambiguous trials can be gated out. Calibrate on BG_046; test transfer to other subjects (features are subject-agnostic proportions).

0 · System architecture & data flow

Amber = your steps (human-in-the-loop). Blue = automated. Green = validation. Dashed amber = the refinement loop (relabel where the tagger disagrees with your eye).



1 · What you commit to when you label

Top: the raster you read (one tick per trial, colored by outcome). Below: three data-model options for painting states. ■ hit ■ miss ■ FA/early-lick ■ abort · tints: green=StimSens, violet=disengaged, red=impulsive

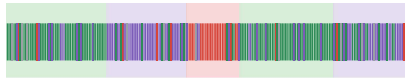
The raster — synthetic ~240-trial session: engaged → disengaged → impulsive burst → engaged → disengaged



trial index →

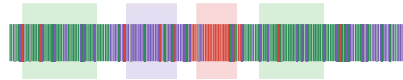
A — Contiguous segmentation

Partition the whole session into back-to-back labeled blocks. Every trial gets a state.



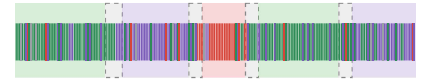
B — Sparse episodes CHOSEN

Paint only confident stretches; leave ambiguous trials blank. Calibration learns from clean exemplars.



C — Hybrid + "uncertain"

Contiguous, but with a hatched uncertain band between confident blocks.



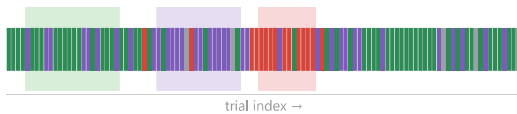
2 · The labeling tool — go vs catch

Same session and same painted episodes; only the raster encoding differs. Rolling-rate curves are kept **OFF** during labeling so labels stay independent of the fractions we fit (avoids circularity).

Color = the lick decision's valence (the ▼ notch separately marks catch trials): ■ green = appropriate lick (hit on a real change) → **StimSens** · ■ red = inappropriate lick — early lick (FA) or lick on a catch trial (SDT false alarm) → **Impulsive** · ■ purple = no lick — missed a real change or correct rejection → **Disengaged** · ■ grey = abort. This maps colors directly onto the three states and prevents catch licks from masquerading as hits.

A — Outcome only

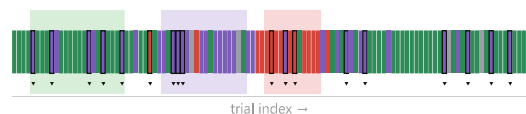
Catch correct-rejections look identical to go-trial misses → can read as "disengaged".



trial index →

B — Go vs catch distinguished CHOSEN

Catch trials get an outline + ▼ notch → withheld catch trials are distinct from go-trial misses.



trial index →

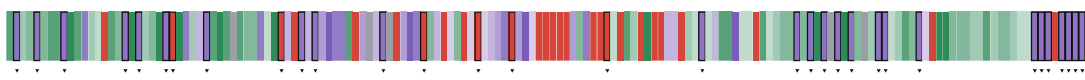
Interaction: drag to paint an episode · keys 1=engaged 2=disengaged 3=impulsive · n=new label · backspace=erase · only painted trials saved · ←/→ navigate sessions (Expert→Naive) · r=toggle rolling overlay (default off).

3 · Optional toggle — shade hits & misses by change size

Go-trial hits become 5 greens and genuine (go-trial) misses become 5 purples by change size; darker = bigger/more obvious change. Difficulty changes how diagnostic an outcome is.

Toggle change-size shading (currently: ON)

BG_046 · 14082025 (Expert)



Hit shades (go trials)



1.25x

1.35x

1.5x

2x

4x

Miss shades (genuine misses)



1.25x

1.35x

1.5x

2x

4x

How to read it: a dark-purple tick = missed an *obvious* change → strong disengagement signal; a light-green tick = caught a *hard* change → strong sensitivity signal. Implication: the calibration features become difficulty-aware (miss-fraction split by easy/hard) so the fitted rule can reproduce a difficulty-weighted eye.

Decisions locked this session

- **Signal:** trial-by-trial outcome raster (local composition), not smoothed curves.

- **Vocabulary:** draft set (StimSens / Disengaged / Impulsive), refined while labeling.
- **Window W:** single fitted value (detection resolution, not a cap on state duration); validation re-fits W per stage to test whether the optimal timescale grows Naive→Expert.
- **Data model:** sparse confident episodes; unlabeled allowed.
- **Calibration:** explicit interpretable thresholds + confidence; fitted window; soft classifier deferred for comparison.
- **Cross-subject:** calibrate on BG_046, test transfer; pool only if transfer fails.
- **Order:** Expert sessions first, then backward to Naive.
- **Display:** go/catch distinguished; rolling curves off by default; optional change-size shading toggle.
- **Color = lick valence:** appropriate lick=green, inappropriate lick (early lick + catch SDT-FA)=red, no lick (miss + correct rejection)=purple, abort=grey — so catch licks read as impulsive, not as hits.